Symbiosis University of Applied Sciences
A Symbiosis Skills University, Indore

School of Computer Science and Technology

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SYNOPSIS REPORT

ON

*"**AIVA:** AI-Based Academic Assistant for Smart Task Management and Study Planning"*

IInd Semester

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1. Introduction

AIVA is an AI-powered academic assistant designed to help students manage tasks, plan studies, and analyze workload efficiently using natural language interaction.

1.1 Problem Statement

Students struggle with managing multiple academic tasks and planning studies efficiently

1.2 Proposed Solution

AIVA provides AI-based task management, study planning, and workload analysis.

2. Objectives

• Smart task creation using AI
• Study planning
• Workload analysis
• Conversational assistant

3. Technology Stack

Backend: FastAPI
Frontend: React
Database: SQLite
AI: OpenAI API

4. Features

• Natural language commands
• Task prioritization
• Study plan generation
• Workload analysis

5. Conclusion

AIVA improves student productivity through intelligent automation and AI-based planning.

1. INTRODUCTION

1.1 Abstract of the Project

In modern academic environments, students face increasing challenges in managing multiple responsibilities such as coursework, competitive exam preparation, projects, and personal development. Traditional task management tools lack intelligence, context-awareness, and adaptability to dynamic academic needs.
This project introduces AIVA (AI Virtual Academic Assistant), an AI-driven platform designed to assist students in managing tasks, planning studies, and optimizing workload using Natural Language Processing (NLP) and intelligent decision-making systems. AIVA enables students to interact conversationally, allowing them to create, update, and manage tasks using natural language commands.
The system integrates AI-based intent recognition, conversational memory, task prioritization, workload analysis, and automated study planning. By leveraging modern AI models and a structured backend architecture, AIVA aims to provide a personalized, efficient, and intelligent academic assistant that enhances productivity and reduces cognitive overload.

1.2 Synopsis of the Project

The project focuses on developing a smart academic assistant platform that combines AI, backend systems, and modern frontend design to provide an integrated student productivity solution.
The core of the system is an AI engine that interprets user input into structured actions such as task creation, updates, deletion, and planning. The system maintains conversation memory, enabling contextual understanding of user commands.
AIVA includes:

- AI-powered task management using natural language
- Intelligent priority scoring based on deadlines and workload
- Daily task recommendation system
- AI-generated study plans with user approval
- Workload analysis to prevent burnout
- Conversational interaction similar to modern AI assistants

The system is built using FastAPI (backend), React (frontend), and OpenAI API (AI layer), ensuring scalability, modularity, and real-world applicability.

1.3 Problem of Existing System

Current student productivity tools face several limitations:

- Lack of Intelligence: Tools like to-do apps do not understand natural language
- No Context Awareness: Cannot handle conversational commands like “make it 3 hours”
- Poor Planning: No automated study planning or workload balancing
- Manual Input Overload: Requires constant manual updates
- No Personalization: Same interface for all users without adaptive suggestions
- No Workload Monitoring: Students often overburden themselves unknowingly

1.4 Solution to the Problem

The proposed system overcomes these limitations by introducing:

- AI-based Natural Language Interface
- Users can interact using simple commands like “Add DSA tomorrow”
- Conversational Memory System
- Understands context like “Make it 3 hours”
- Priority Scoring Engine
- Automatically ranks tasks based on urgency and effort
- AI Study Planner
- Generates structured study plans for multiple days
- Workload Analyzer
- Evaluates weekly workload and provides feedback
- Dynamic Task Management
- Supports creation, update, deletion using AI

2. PROJECT INTRODUCTION

2.1 Functional Requirements

- FR1: AI-based task creation using natural language
- FR2: Task update and deletion using conversational commands
- FR3: Daily task recommendation system
- FR4: AI-generated study planning with approval workflow
- FR5: Workload analysis and overload detection
- FR6: Conversation memory for contextual understanding
- FR7: Deadline parsing for natural date formats

2.2 Non-Functional Requirements

- Performance: Fast response time for AI queries
- Scalability: Supports increasing number of tasks and interactions
- Usability: Clean, minimal UI for students
- Reliability: Accurate AI intent detection
- Maintainability: Modular backend architecture
- Security: Safe API handling and data storage

2.3 Process Model

The system follows this workflow:

1. User Input (Natural Language)
2. AI Processing (Intent Detection via OpenAI)
3. JSON Conversion (Structured Command)
4. Backend Execution (Task Service)
5. Database Storage (SQLite)
6. Response Generation
7. Frontend Display

3. SOFTWARE REQUIREMENT SPECIFICATION

3.1 Software Requirements

Programming Languages

- Python (Backend)
- JavaScript (Frontend)

Frameworks & Libraries

- FastAPI (Backend API)
- React (Frontend UI)
- SQLAlchemy (Database ORM)
- OpenAI API (AI Processing)
- Tailwind CSS (UI Styling)

Database

- SQLite (Prototype)

Tools

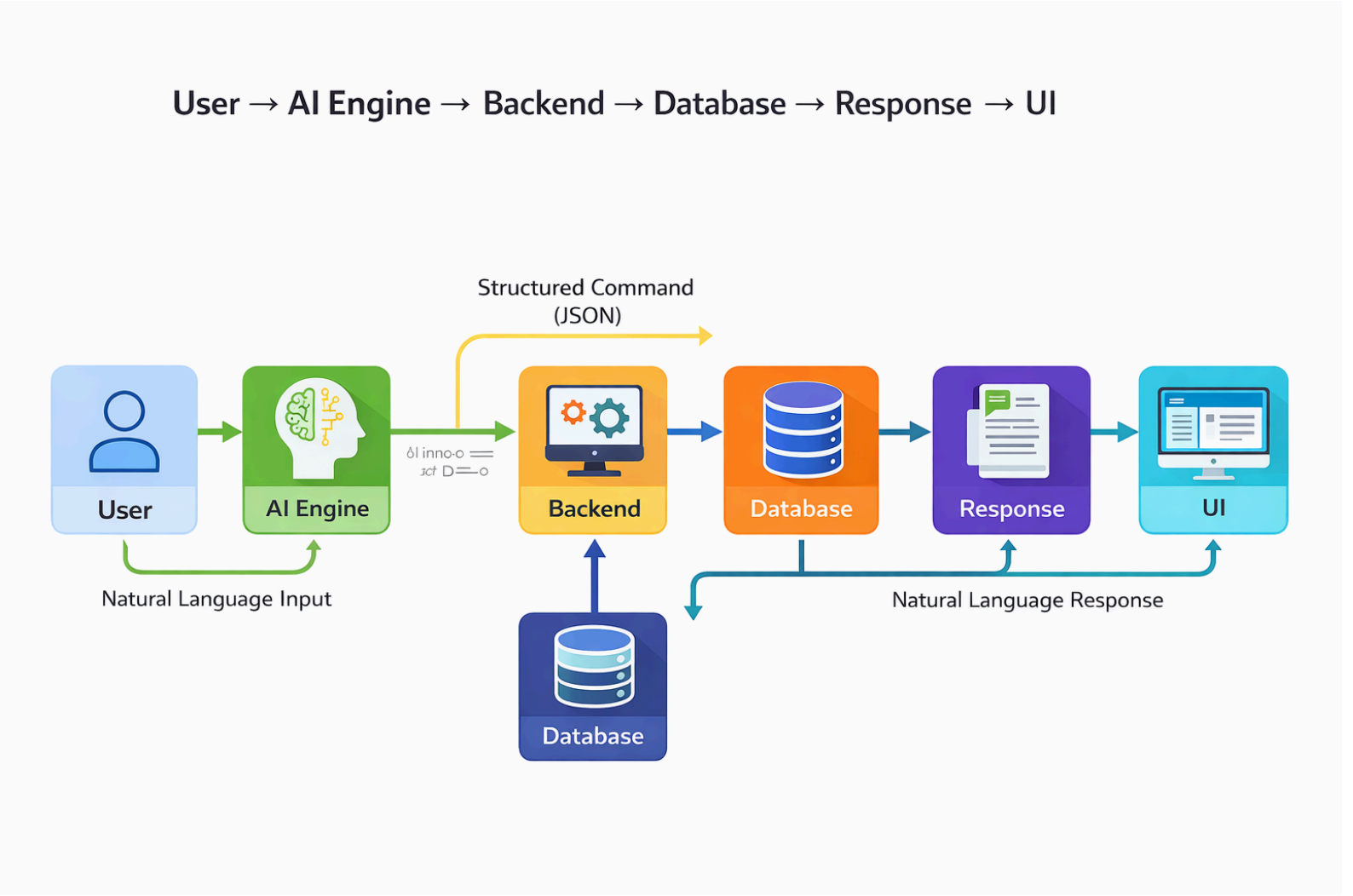
- VS Code
- GitHub

Operating System

- Windows / Linux

4. SYSTEM ANALYSIS

4.1 Data Flow Diagram (Conceptual)



Components:

- AI Engine (Intent detection)
- Task Service (Execution logic)
- Database (Task storage)
- Frontend (User interface)

5. CONCLUSION

The proposed system, AIVA (AI Virtual Academic Assistant), provides an intelligent and user-friendly solution for managing academic tasks and study planning. By integrating AI-driven natural language understanding, conversational memory, and smart workload analysis, the system significantly improves student productivity and reduces manual effort.

Unlike traditional task management systems, AIVA offers context-aware interaction, dynamic planning, and intelligent recommendations, making it a powerful academic companion. The modular design ensures scalability and future enhancements such as voice interaction, advanced analytics, and cross-platform deployment.

In conclusion, AIVA represents a modern approach to student productivity, combining AI and software engineering to create a smart, adaptive, and efficient academic assistant.